DSO 576 - Module 6 Project Data Cleaning

Emerging Artist Discovery Through Listening Behavior

Student	Shivanee Kooner
Branch	cleaning-shivanee
Branch URL	https://github.com/jennzhong/dso576-group-project/tree/cleaning-shivanee
Submitted cleaning commit	7f4a60bcecc30a2480348c1c451f0dd8487b7739
Data source	ListenBrainz public API
Historical cutoff	2026-09-22 00:00:00 UTC (includes listens through Sep. 21, 2026)

Project question

Can listening behavior - especially growth in unique listeners and repeat engagement - help a music-streaming editorial team identify promising emerging artists for additional exposure?

One row

One listen event: one user playing one track at one point in time.

14-item checklist

1. State the project vision and question

Status: Done

Evidence: Project vision: a music-streaming editorial team uses listener growth and repeat engagement to shortlist emerging artists. Business question: Can listening behavior - especially growth in unique listeners and repeat engagement - help a music-streaming editorial team identify promising emerging artists for additional exposure? Main comparison: artist-level unique-listener growth and repeat engagement over time.

2. Define one row

Status: Done

Evidence: One row represents one listen event: one user playing one track at one point in time. Audit event key: ['username', 'listened_at', 'recording_msid']. Repeated track or artist plays are legitimate behavior and are not automatically duplicates.

3. Preserve the source

Status: Done

Evidence: Source: ListenBrainz public API, historical cutoff 2026-09-22 00:00:00 UTC, recreated with pull_listenbrainz_sample.py. Raw file listenbrainz_sample.csv is never overwritten. Cleaning output is written separately.

4. Inspect columns and types

Status: Done

Evidence: Expected nine columns were checked. listened_at was converted to UTC datetime. Failed date conversions: 0. Text and identifier fields remain text; artist_mbids remains a serialized list value.

5. Standardize text

Status: Done

Evidence: Trimmed only leading/trailing whitespace in text and identifier fields. Preserved case, punctuation, accents, and internal spacing because these can be meaningful in artist and track names. artist_mbids was normalized to compact JSON list text.

6. Investigate duplicates

Status: Done

Evidence: Raw exact duplicate copies: 0. Normalized exact duplicate copies removed: 0. Rows involved in repeated event keys after cleaning: 0. Only exact duplicates were removed; repeated listening was retained.

7. Handle missing data

Status: Done

Evidence: Missing counts before: {'username': 0, 'listened_at': 0, 'track_name': 0, 'artist_name': 0, 'release_name': 754, 'recording_msid': 0, 'artist_mbids': 759, 'recording_mbid': 728, 'release_mbid': 759}. Missing counts after: {'username': 0, 'listened_at': 0, 'track_name': 0, 'artist_name': 0, 'release_name': 754, 'recording_msid': 0, 'artist_mbids': 759, 'recording_mbid': 728, 'release_mbid': 759}. Missing MusicBrainz identifiers and release names were not imputed because guessing could create incorrect joins. Core ListenBrainz event fields were checked separately.

8. Check suspicious values

Status: Done

Evidence: Invalid/failed dates: 0; dates at or after historical cutoff: 0; core-field missing counts: {'username': 0, 'track_name': 0, 'artist_name': 0, 'recording_msid': 0}. Unusual values are flagged rather than silently rewritten.

9. Verify merges

Status: Not applicable

Evidence: No table merge is performed in this Module 6 cleaning step. The project plans a later MusicBrainz enrichment using artist_mbids -> MusicBrainz artist MBID. This assignment cleans the ListenBrainz event table first.

10. Reconcile changes

Status: Done

Evidence: Starting rows: 1940; final rows: 1940; difference: 0. The difference is explained by 0 exact duplicate copies removed. Unique listeners after cleaning: 5; unique artist names: 756.

11. Check calculations

Status: Done

Evidence: Audit groupby with dropna=False reconciles to 1940 listens versus 1940 final rows. Resolved artist-MBID rows: 1181 of 1940. MBID-based metrics must state that unresolved rows are excluded; overall row/listener audits include all rows.

12. Verify individual records

Status: Done

Evidence: Five difficult/representative records are documented in record_checks.csv. All expected-versus-actual checks match: True.

13. Test reproducibility

Status: Done

Evidence: Reproduction command: install requirements, run pull_listenbrainz_sample.py, then clean_listenbrainz.py, then build_report.py; alternatively Run All in module6_cleaning.ipynb. The scripts rebuild the raw extract, cleaned evidence files, 50-row sample, and report from source.

14. Document remaining limitations

Status: Done

Evidence: The sample contains only five self-selected ListenBrainz users, so listener counts are sample-specific. User observation windows are uneven and do not share one common time period. The historical extract is capped for several users, so it is not each user's complete listening history. MusicBrainz identifiers are missing for some listens, and the missingness is uneven across users. A listener's first play in this extract may not be their true first exposure to an artist. The data has no listener demographics, skip signal, or source-of-discovery field, so segmentation is behavioral only. ListenBrainz server-side MBID resolution and historical records can change after the original September 2026 project extract.

Cleaning decision log

Rule	Reason	Check
Preserve raw source	The assignment requires the original data to remain unchanged.	Cleaning reads listenbrainz_sample.csv and writes a separate listenbrainz_cleaned.csv.
Parse listened_at as UTC datetime	Growth and engagement depend on correct time ordering and time windows.	Failed date conversions after parsing: 0.
Trim outer whitespace only in text/ID fields	Accidental edge whitespace can split equivalent values, but case, punctuation, accents, and internal spacing may be meaningful artist/track metadata.	Compared before/after values in the five record checks; no forced lowercasing or punctuation removal.
Normalize artist_mbids as a JSON list string	Artist MBIDs are list-valued and should have a reproducible representation without changing artist-credit order.	Parsed JSON/Python-list forms and re-serialized valid nonempty lists; missing/unparseable values remain missing.
Remove exact duplicate rows only	Repeated listening is a real behavior in this project, so repeated users/artists/tracks must not be removed merely because they repeat.	Raw duplicate copies: 0; normalized duplicate copies removed: 0; repeated event-key rows flagged separately: 0.
Do not impute missing MusicBrainz identifiers or release names	Missing MBID mappings mean ListenBrainz did not resolve the listen. Guessing an identifier could create incorrect artist joins and bias the analysis.	Resolved artist-MBID rows after cleaning: 1181 of 1940; missing counts are reported before and after.
Flag suspicious core values instead of silently rewriting them	Unusual does not automatically mean wrong. Invalid dates or missing core fields would affect the analysis and should remain auditable.	Dates at/after historical cutoff: 0; core-field missing counts: {'username': 0, 'track_name': 0, 'artist_name': 0, 'recording_msid': 0}.
No MusicBrainz metadata merge in this Module 6 cleaning step	The current input is the ListenBrainz event table; the planned MusicBrainz enrichment is a later project step.	Checklist item 9 is marked Not applicable and explains the planned MBID join separately.

Five record checks

The expected result was defined from the written cleaning rule before it was compared with the actual cleaned output. The full original/expected/actual values are also saved in record_checks.csv.

Record check 1

Case	Missing artist MBID preserved
Source row	31
Original artist	Sachin-Jigar, Arijit Singh, Amitabh Bhattacharya
Expected artist	Sachin-Jigar, Arijit Singh, Amitabh Bhattacharya
Actual artist	Sachin-Jigar, Arijit Singh, Amitabh Bhattacharya
Original time	2024-12-16 03:37:09
Expected time	2024-12-16T03:37:09+00:00
Actual time	2024-12-16T03:37:09+00:00
Original artist_mbids	nan
Expected artist_mbids	nan
Actual artist_mbids	nan
Matches expected?	Yes

Record check 2

Case	Missing release name preserved
Source row	500
Original artist	Pills
Expected artist	Pills
Actual artist	Pills
Original time	2019-11-15 18:03:47
Expected time	2019-11-15T18:03:47+00:00
Actual time	2019-11-15T18:03:47+00:00
Original artist_mbids	nan
Expected artist_mbids	nan
Actual artist_mbids	nan
Matches expected?	Yes

Record check 3

Case	Multiple artist MBIDs preserved in credit order
Source row	0
Original artist	Drake, Rihanna
Expected artist	Drake, Rihanna
Actual artist	Drake, Rihanna
Original time	2024-12-16 08:04:37
Expected time	2024-12-16T08:04:37+00:00
Actual time	2024-12-16T08:04:37+00:00
Original artist_mbids	['9fff2f8a-21e6-47de-a2b8-7f449929d43f', '73e5e69d-3554-40d8-8516-00cb38737a1c']
Expected artist_mbids	["9fff2f8a-21e6-47de-a2b8-7f449929d43f","73e5e69d-3554-40d8-8516-00cb38737a1c"]
Actual artist_mbids	["9fff2f8a-21e6-47de-a2b8-7f449929d43f","73e5e69d-3554-40d8-8516-00cb38737a1c"]
Matches expected?	Yes

Record check 4

Case	Fallback record check
Source row	1
Original artist	Jungle
Expected artist	Jungle
Actual artist	Jungle
Original time	2024-12-16 05:56:01
Expected time	2024-12-16T05:56:01+00:00
Actual time	2024-12-16T05:56:01+00:00
Original artist_mbids	['6bbb3983-ce8a-4971-96e0-7cae73268fc4']
Expected artist_mbids	["6bbb3983-ce8a-4971-96e0-7cae73268fc4"]
Actual artist_mbids	["6bbb3983-ce8a-4971-96e0-7cae73268fc4"]
Matches expected?	Yes

Record check 5

Case	Fallback record check
Source row	2
Original artist	Cannons
Expected artist	Cannons
Actual artist	Cannons
Original time	2024-12-16 05:44:12
Expected time	2024-12-16T05:44:12+00:00
Actual time	2024-12-16T05:44:12+00:00
Original artist_mbids	['eb9eb9f8-c691-4fef-9bdf-0cbffbfa42ee']
Expected artist_mbids	["eb9eb9f8-c691-4fef-9bdf-0cbffbfa42ee"]
Actual artist_mbids	["eb9eb9f8-c691-4fef-9bdf-0cbffbfa42ee"]
Matches expected?	Yes

How one cleaning decision was checked

I did not use Codex. I checked the decision to remove only exact duplicate rows by reconciling the starting row count, the number of duplicate copies removed, and the final row count. I also verified five original records against rule-based expected values and the actual cleaned output. Repeated artist or track listens were retained because one row represents a listen event and repeat listening is part of the project behavior.

50-row cleaned sample

The 50-row sample intentionally includes the five record-check rows, oldest and newest listens for users where available, rows with missing artist MBIDs, multi-artist-credit cases when present, and a deterministic random sample (random_state=576) to reach 50 rows.

The full 50-row sample is included in the GitHub branch as cleaned_sample_50.csv.

Remaining limitations

- The sample contains only five self-selected ListenBrainz users, so listener counts are sample-specific.

- User observation windows are uneven and do not share one common time period.

- The historical extract is capped for several users, so it is not each user's complete listening history.

- MusicBrainz identifiers are missing for some listens, and the missingness is uneven across users.

- A listener's first play in this extract may not be their true first exposure to an artist.

- The data has no listener demographics, skip signal, or source-of-discovery field, so segmentation is behavioral only.

- ListenBrainz server-side MBID resolution and historical records can change after the original September 2026 project extract.